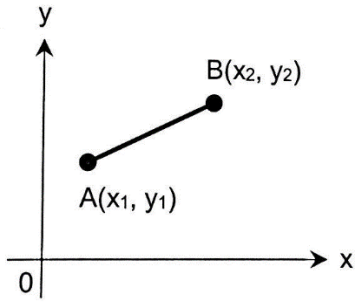
Chapter 1

1.4 Coordinate Geometry

Distance between Two Points



$\text{Distance AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Example 1

Find the distance between the points P(-2, 7) and Q(4, -1).

$$\begin{aligned}\text{Distance PQ} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(4 - (-2))^2 + (-1 - 7)^2} \\ &= \sqrt{(6)^2 + (-8)^2} \\ &= \sqrt{100} \\ &= 10 \text{ units}\end{aligned}$$

Example 2

Given the points A(8, k), B(-2,1) and C(3,2). If the distance AB is twice of the distance BC. Find the possible values of k.

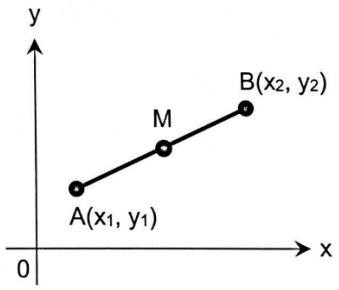
Solution

$$AB = 2BC$$
$$\sqrt{(-2-8)^2 + (1-k)^2} = 2\sqrt{[3-(-2)]^2 + (2-1)^2}$$
$$\sqrt{(-10)^2 + (1-k)^2} = 2\sqrt{5^2 + 1^2}$$

Square both sides of the equation:

$$\left[\sqrt{(-10)^2 + (1-k)^2}\right]^2 = \left[2\sqrt{5^2 + 1^2}\right]^2$$
$$(-10)^2 + (1-k)^2 = 4(25+1)$$
$$100 + 1 - 2k + k^2 = 104$$
$$k^2 - 2k + 101 = 104$$
$$k^2 - 2k - 3 = 0$$
$$(k-3)(k+3) = 0$$
$$k = 3 \quad \text{or} \quad k = -1$$

Midpoint between Two Points



$$\text{Midpoint of AB} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example 3

Find the midpoint of a straight line joining the points A(1, 6) and B(7, 4).

$$\begin{aligned}\text{Midpoint of AB} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)\\ &= \left(\frac{1+7}{2}, \frac{6+4}{2}\right)\\ &= \left(\frac{8}{2}, \frac{10}{2}\right)\\ &= (4, 5)\end{aligned}$$

Example 4

Given F(1, -5) is the value of midpoint of the points E(m, 2) and G(-3, n). Find the values of m and n.

Solution

$$\text{Midpoint of EG} = (1,-5)$$

$$\left(\frac{m + (-3)}{2}, \frac{2+n}{2}\right) = (1, -5)$$

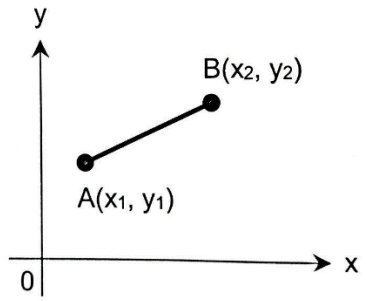
$$\left(\frac{m-3}{2}, \frac{2+n}{2}\right) = (1, -5)$$

$$\frac{m-3}{2} = 1 \;\; ; \;\; \frac{2+n}{2} = -5$$

$$m - 3 = 2 \;\; ; \;\; 2 + n = -10$$

$$m = 5 \;\; ; \;\; n = -12$$

Gradient of Straight Line



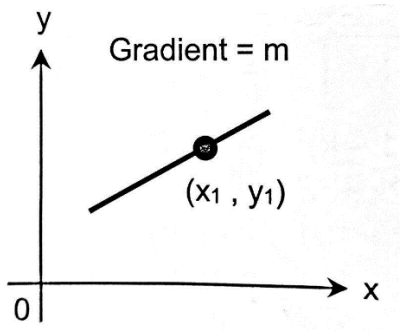
Gradient of AB, $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$

Example 5

Find the gradient of the straight line which passes through P(3, 1) and Q(-1, 9).

$$m_{PQ} = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{9 - 1}{-1 - 3}$$
$$= \frac{8}{-4}$$
$$= -2$$

Equations of Straight Line



The equation of a straight line which has a gradient, m and passes through the points (x, y) is give by $y = mx + c$ or $y - y_1 = m(x - x_1)$.

Example 6

Find the equation of a straight line which has gradient of -2 and passes through the point (-1,4)

Method 1

$y = mx + c$
$4 = -2(-1) + c$
$4 = 2 + c$
$c = 2$
$\therefore y = -2x + 2$

Method 2

$y - y_1 = m(x - x_1)$
$y - 4 = -2(x - (-1))$
$y - 4 = -2(x + 1)$
$y - 4 = -2x - 2)$
$y = -2x - 2 + 4$
$y = -2x + 2$

Finding the Gradient from the Equations of Straight Line

Example 7

Find the gradient for each of the following straight lines.
a) $y = 3x - 2$
b) $x + y = 2$
c) $2x + 4y = 5$

Solution

a) Equation $y = 3x - 2$ is compared to $y = mx + c$, then $m = 3$

b) Write equation $x + y = 2$ in the form of $y = mx + c$
$x + y = 2$
$y = -x + 2$
$\therefore\ m = -1$

c) Write equation $2x + 4y = 5$ in the form of $y = mx + c$
$2x + 4y = 5$
$4y = -2x + 5$
$y = \frac{-2}{4}x + \frac{5}{4}$
$y = -\frac{1}{2}x + \frac{5}{4}$
$\therefore\ m = -\frac{1}{2}$

Intersection Point of Two Lines

Given the straight lines $2x + y = 4$ and $x - y = -1$ intersect at the point A. Find the coordinates of A.

$$2x + y = 4. \ldots \ldots \quad \ldots \ldots (1)$$
$$x - y = -1. \ldots \ldots \ldots \ldots (2)$$

(1) + (2) $3x \quad = 3$

$$x = 1$$

Substitute $x = 1$ into equation (2)

$1 - y = -1$

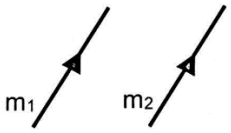
$-y = -2$

$y = 2$

$\therefore A = (1, 2)$

Parallel Lines and Perpendicular Lines

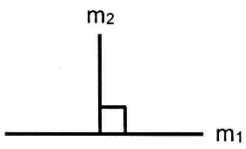
a) Parallel Lines



If two straight lines having gradients m_1 and m_2 respectively are **parallel**, then

$m_1 = m_2$

b) Perpendicular Lines



If two straight lines having gradients m_1 and m_2 respectively are **perpendicular**, then

$m_1 \times m_2 = -1$

Example 8

Find the equation of the straight line which passes through the point $A(-2,5)$ and is parallel to the line $6x - 3y + 2 = 0$.

Solution

$-3y = -6x - 2$
$y = \frac{-6x}{-3} - \frac{2}{-3}$
$y = 2x + \frac{2}{3}$ $\boxed{m_1 = 2}$

Since the two lines are parallel, then $m_1 = m_2$

So $m = 2$, $A(-2,5)$
$y = mx + c$
$5 = 2(-2) + c$
$c = 9$
$\therefore y = 2x + 9$

Example 9

Find the equation of the straight line which passes through the point $Q(3,-4)$ and is perpendicular to the line $y + 5x - 3 = 0$.

Solution

$y = -5x + 3$
$m_1 = -5$

$-5 \times m_2 = -1$
$m_2 = \frac{1}{5}$

So $m = \frac{1}{5}$, $Q(3,-4)$
$y = mx + c$
$-4 = \frac{1}{5}(3) + c$
$c = -\frac{23}{5}$
$\therefore y = \frac{1}{5}x - \frac{23}{5}$